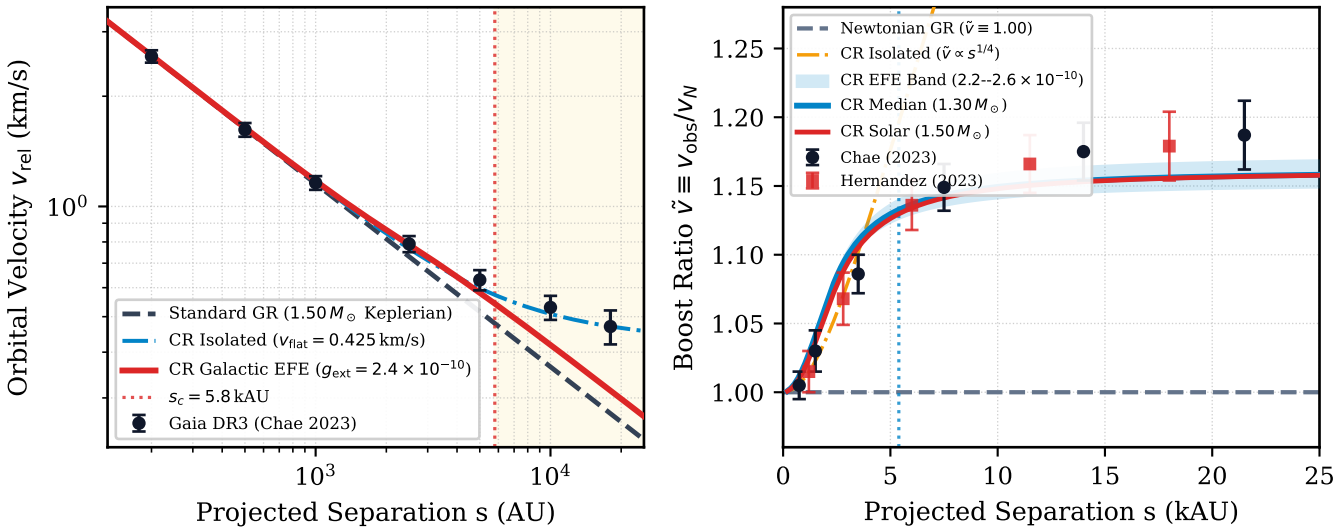
(a) Solar-Type Kinematics (1.50 $M_\odot$)
Orbital Velocity v_{rel} (km/s)
10^0
10^3
10^4
Projected Separation s (AU)
Standard GR (1.50 $M_\odot$ Keplerian)
CR Isolated ($v_{flat} = 0.425$ km/s)
CR Galactic EFE ($g_{ext} = 2.4 \times 10^{-10}$)
$s_c = 5.8$ kAU
Gaia DR3 (Chae 2023)
(b) Normalized Velocity Boost $\tilde{v}$
Boost Ratio $\tilde{V} \equiv v_{obs}/v_N$
1.25
1.20
1.15
1.10
1.05
1.00
0
5
10
15
20
25
Projected Separation s (kAU)
Newtonian GR ($\tilde{v} \equiv 1.00$)
CR Isolated ($\tilde{v} \propto s^{1/4}$)
CR EFE Band (2.2--2.6 $\times 10^{-10}$)
CR Median (1.30 $M_\odot$)
CR Solar (1.50 $M_\odot$)
Chae (2023)
Hernandez (2023)